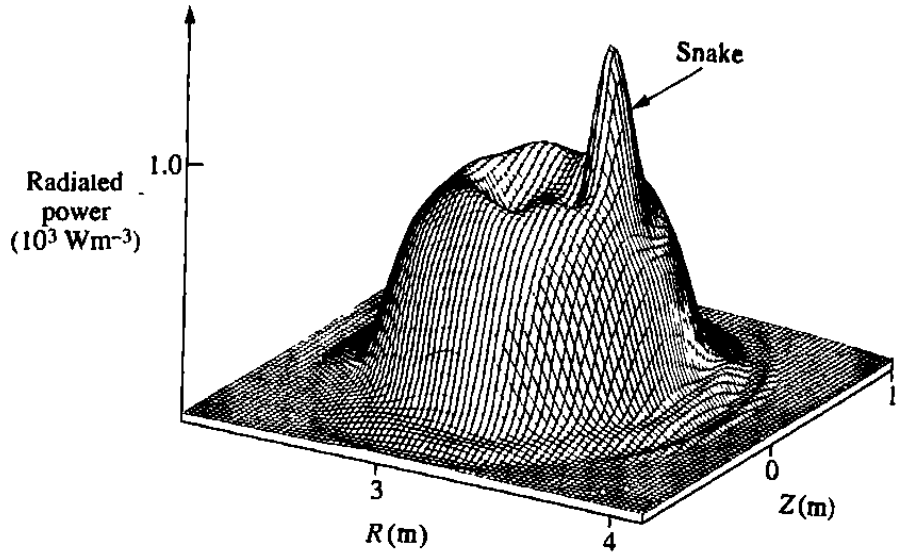


Fig. 10.8.7 Tomographic reconstruction of the snake which is a small region of enhanced plasma density in JET produced on the $q = 1$ magnetic surface following injection of a solid D_2 pellet.

10.9 Langmuir probes

In its simplest form the Langmuir probe consists of a single electrode which can be inserted into the plasma. A voltage is applied to the probe with respect to the vessel and the current is measured. From the current–voltage (I–V) characteristic of such a probe the electron temperature T_e and density n_e can be deduced. In large tokamaks the use of Langmuir probes is limited to the plasma edge because of the large power fluxes in the interior.

The interpretation of Langmuir probes is based on the theory of the electrostatic sheath (see Section 9.2) which is established at the surface of the probe element. Combination of eqns 9.2.8 and 9.2.9 leads to an expression for the current to a probe I which is biased to voltage V

$$I = (j_i + j_e)A = j_i(1 - \exp(e(V - V_f)/T_e))A, \qquad 10.9.1$$

where j_i and j_e are the electron and ion current densities and A is the projected surface area of the probe. In the limit where the probe dimensions are large compared to the ion and electron gyro-radii and the Debye length, $A = \int \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{S}/\mathrm{B}$. V_f is the floating potential discussed below. When the probe is biased sufficiently negatively all electrons are repelled and all that remains is the ion current. This current is independent of voltage and is called the ion saturation current. Saturation of the ion current occurs because all the bias potential appears across the thin shesth and the ion flux is simply determined by the flux of ions crossing the sheath edge at the ion sound speed.

The floating potential V_f is the potential at which the probe would float if it were electrically isolated. If an external potential is applied to

the probe then V_f is determined by the applied potential at which the net current is zero. As the probe voltage is swept positive with respect to V_f, it is seen from eqn 10.9.1 that the electron current rises exponentially. However, eqn 10.9.1 is only applicable when the probe potential lies below the plasma potential. When the probe potential equals the plasma potential there is no sheath electric field and the electrons flow to the probe surface with a thermal velocity distribution. This point in the characteristic is called electron saturation, and the electron saturation current is given by $I_{es} = \frac{1}{4} n_0 e \bar{c}_e A$, where $\bar{c}_e$ is the electron thermal speed. In non-magnetized plasmas, langmuir probe characteristics often follow this ideal picture with electron to ion saturation current ratios in hydrogen plasmas around 50:1, as expected from the ratio of electron thermal speed to ion sound speed.

Figure 10.9.1 shows an I–V characteristic obtained from a single Langmuir probe in the T10 tokamak. The data is fitted up to the point where the roll over into electron saturation occurs using eqn 10.9.1. In this case the electron to ion saturation current ratio is 6.5 hut ratios as low as 1:1 are sometimes observed. As yet there is no adequate theory for this behaviour and for this reason it is usually advisable to ignore data from the net electron collecting side of the characteristic. The consequence of this is that usually only the tail of the electron distribution, comprising around 5% of the total, is measured.

Assuming equal ion and electron temperatures, the electron density at the sheath edge may be derived from eqn 9.2.8 to give

$$n_0 = \frac{J_i}{e}\left(\frac{m_i}{2T_e}\right)^{1/2}. \qquad 10.9.2$$

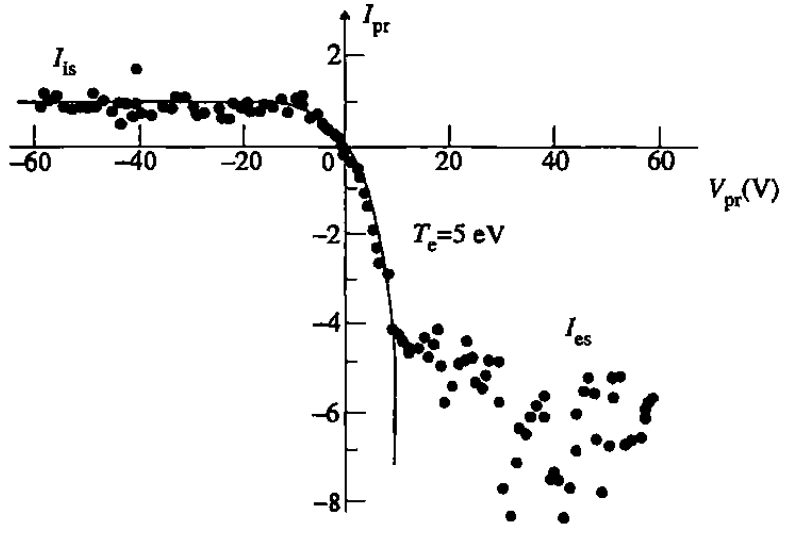


Fig. 10.0.1 Single Langmuir probe characteristic from the T10 tokamak showing low electron to inn saturation current ratio as compared with the vatue predicated by eqn 10.9.1.

Probe assemblies

The two main ways that probes of various types can be introduced into a tokamak are illustrated in Fig. 10.9.2:

1. Arrays of fixed probes can be built into limiters and divertor targets with the purpose of characterizing the plasma contacting the material surfaces. Since such probes do not modify the sheath already existing at the limiter or divertor targets, it is normal to quote the density at the sheath edge, given by eqn 10.9.2.

2. Probes can be mounted on drives which can map out a profile of the edge plasma on a pulse by pulse basis (slow drive) or during a pulse (fast reciprocating drive). Fast drives are essential in large tokamaks where prolonged exposure of the Langmuir probe head leads to vaporization. Probes of this type act as sinks for plasma ions and so locally perturb the plasma. This density perturbation occurs over a distance called the collection length $L_{\rm col}$ which is determined by the rate at which the parallel flow of ions to the probe can be supplied by cross-field diffusion

$$L_{\rm col} \simeq \frac{d^2 c_{\rm s}}{D_\perp},$$

where d is the projected probe width, $c_{\rm s}$ the ion sound speed and $D_\perp$ the diffusion coefficient perpendicular to the magnetic field. Provided that $L_{\rm col}$ is less than the distance along $\boldsymbol{B}$ to the next material object the unperturbed density can be obtained by multiplying the density at the sheath edge, given by eqn 10.9.2, by $\sim$2. This factor allows for the drop in density which occurs when the ions are accelerated up to the ion sound speed at the edge of the sheath created by the probe.

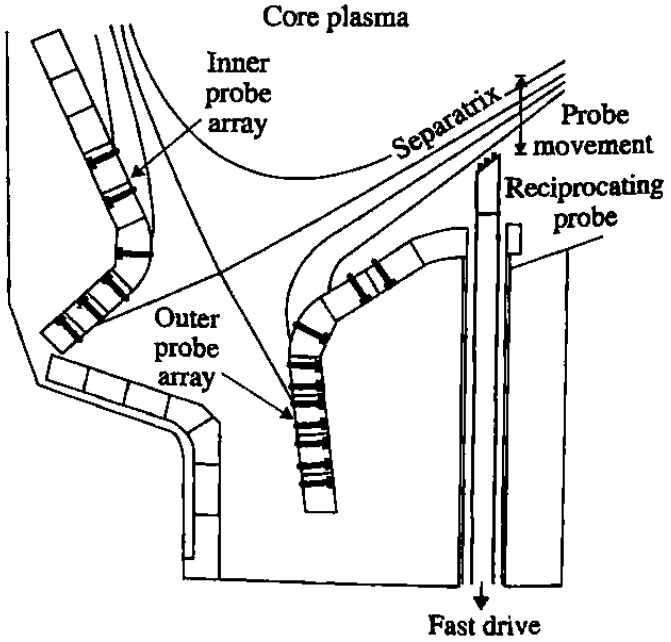


Fig. 10.9.2 Fixed probe arrays and fast reciprocating probe in the divertor region of C-mod tokamak.

Single probes

When current is drawn from a plasma through a probe tip an equal current of opposite sign must be returned to the plasma through a second electrode. In the case of single Langmuir probes in tokamaks, the torus is often taken as the reference electrode. The main requirement is for a surface area sufficiently large that a negligible potential change across the reference electrode sheath is needed for the return current to flow. This requires that the surface carrying the return current does not reach the ion-saturation limit before the probe reaches electron saturation. In a uniform hydrogenic plasma the reference electrode would need an area which is larger than the probe area by a factor in excess of the ideal electron to ion saturation current ratio, which is typically ~50:1. Despite the large surface area within a tokamak, currents flow much more readily along the magnetic field than across it and so the area of surface from which the return current flows may actually be very restricted. This restriction of the current channel has been put forward as a possible explanation for the low electron to ion saturation current ratios which have been observed in many tokamaks and can be seen in the example of Fig. 10.9.1.

Double probes

Double Langmuir probes consist of a pair of probe tips locate close enough together that they can be considered to be exposed to the same plasma conditions. A swept isolated power supply drives current in one tip and out the other so that $I_1 + I_2 = 0$. Defining I_1 and I_2 with eqn 10.9.1 and the power supply voltage as $V_s = V_1 - V_2$, the following equation is obtained for an asymmetric double probe characteristic

$$I_1 = \frac{j_i A_1 (1 - \exp(eV_s/T_e))}{1 + (A_1/A_2)\exp(eV_s/T_e)}.$$

If the areas are equal, that is $A_1 = A_2$ then

$$I_1 = j_i A_1 \tanh(V_s/2T_e).$$

The only advantage that a double probe would appear to have over a single probe is that the return path for the current can be less ambiguous. In practice however, no real benefits have been demonstrated which would justify the added complexity and loss of information about the electron collecting side of the characteristic.

Triple probes

Triple Langmuir probes consists of three probe tips each of which must be exposed to the same plasma parameters. One tip provides the floating potential whilst the other two are differentially biased with a constant potential as an isolated pair such that one tip draws an ion saturation current $j_i A_1$ and the other a net electron current $-j_i A_1$. This arrangement

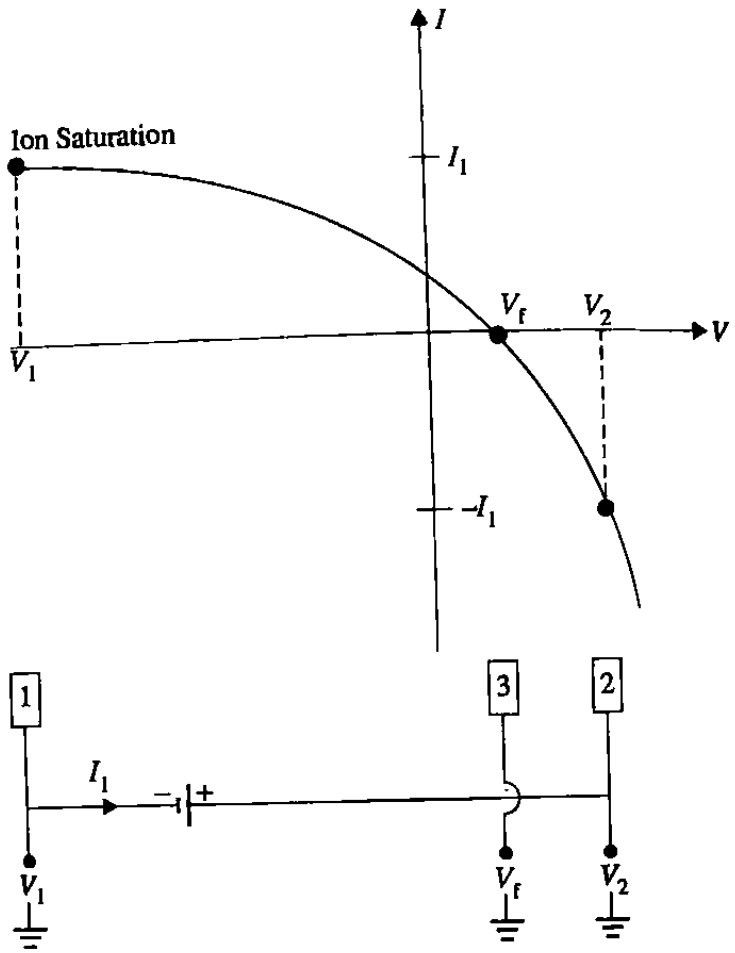


Fig. 10.9.3 Schematic illustrating the principle of the triple Langmuir probe. The circuit diagram is shown together with the positions of the three probes on the $I(V)$ curve.

is shown in Fig. 10.9.3. The potential of the electron collecting side of the circuit V_2 adjusts itself such that the electron side draws an equal but opposite current to the ion side. Applying eqn 10.9.1 to tips 1 and 2 and using the continuity equation $I_1 + I_2 = 0$, the following equation is obtained

$$(1 - e^{e(V_1 - V_f)/T_e})A_1 + (1 - e^{e(V_2 - V_f)/T_e})A_2 = 0. \qquad 10.9.3$$

In the limit where the supply voltage is large $e|V_1 - V_f| \gg T_e$, eqn 10.9.3 gives a simple expression for the electron temperature,

$$T_e = (V_2 - V_f)/\ln\left(1 + \frac{A_1}{A_2}\right).$$

The advantage of the triple probe method is high time resolution since V_2, V_f, and j_i can be measured simultaneously. This contrasts with swept probes where the current and voltage may need to be recorded at up to 100 time points to define a single characteristic from which averaged parameters may be extracted. Figure 10.9.4 shows an example of high time resolution divertor triple probe measurements from an ELMy discharge in JET. Triple probe data is however unreliable under conditions where the ion and electron saturation currents are comparable, or the plasma parameters vary across the three tips.

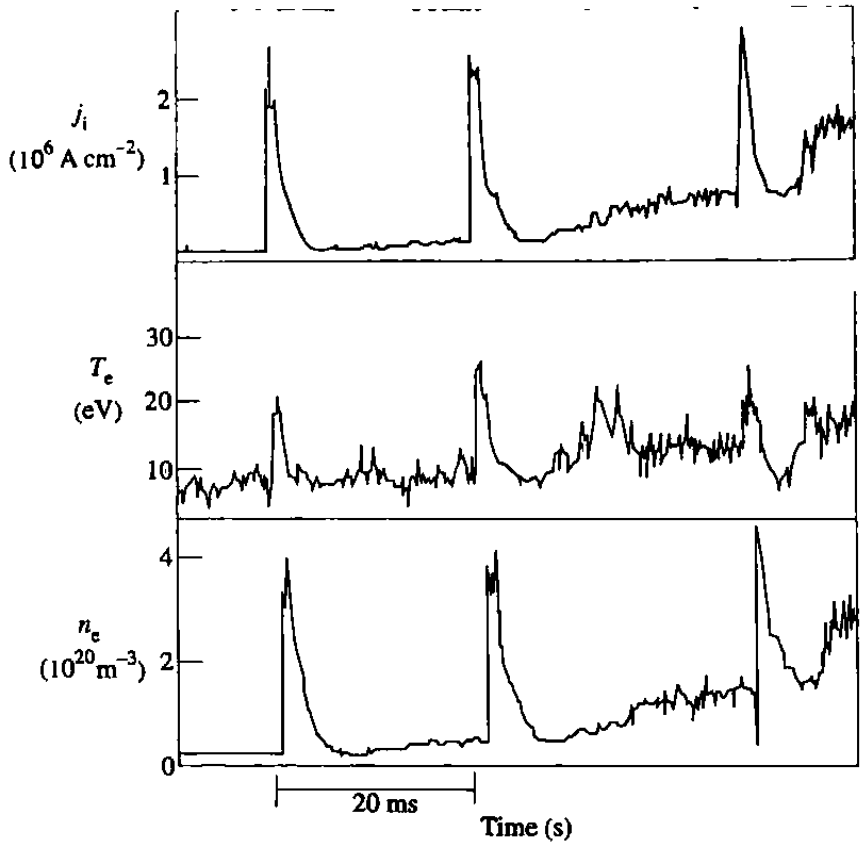


Fig. 10.9.4 Divertor triple probe data from an ELMy discharge in JET.

10.10 Measurements of fluctuations

The experimentally determined energy and particle transport coefficients in a tokamak greatly exceed those calculated from collision theory. It is generally accepted that plasma turbulence is responsible for this anomaly and a considerable experimental effort has been made to try to understand the mechanisms operating, to distinguish between different theoretical models, and to try to show that the experimentally determined turbulence does indeed account for the observed transport.

Electrostatic fluctuations affect both particle transport and heat conduction. The particle flux may be written as

$$\Gamma_j = \frac{\langle \tilde{E}_\perp \tilde{n}_j \rangle}{B},$$

where B is the magnetic field, $\tilde{E}_\perp$ and $\tilde{n}_j$ are the fluctuating parts of the electric field and the particle density respectively. The triangular brackets denote averaging. The heat flux is also affected and the turbulence gives